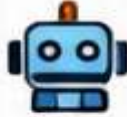


Secure Access Through Biometric IoT

Advanced Biometric Authentication & Cloud Monitoring

PROJECT TITLE	TEAM MEMBERS	COURSE NAME	INSTRUCTOR
Secure Access Through Biometric IoT	Alaa elrawy (2023041831) Malak Mohamed (202304215) Roaa Sabry (203304078) Ritaj Mohamed (202304621)	 Robotics	Hesham Ahmed

Watch Project Video:

Smart Door Demo



Watch Video



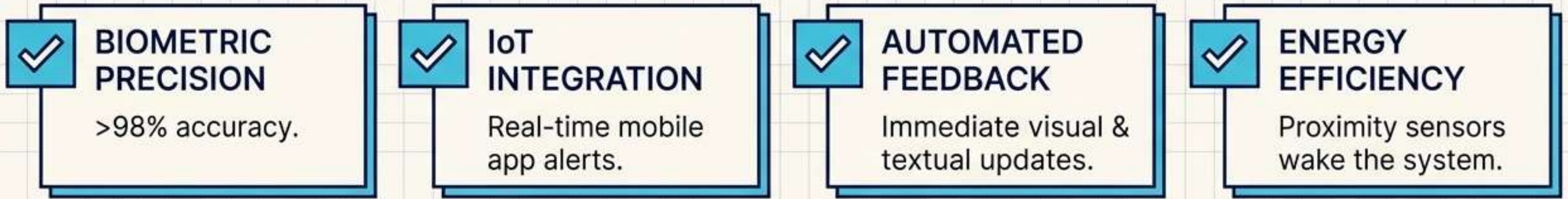
The Evolution of Access

Transitioning from vulnerable traditional keys to the security of **Biological Identity**. Integrating the **ESP32 Microcontroller** bridges the gap between physical locks and real-time digital monitoring.

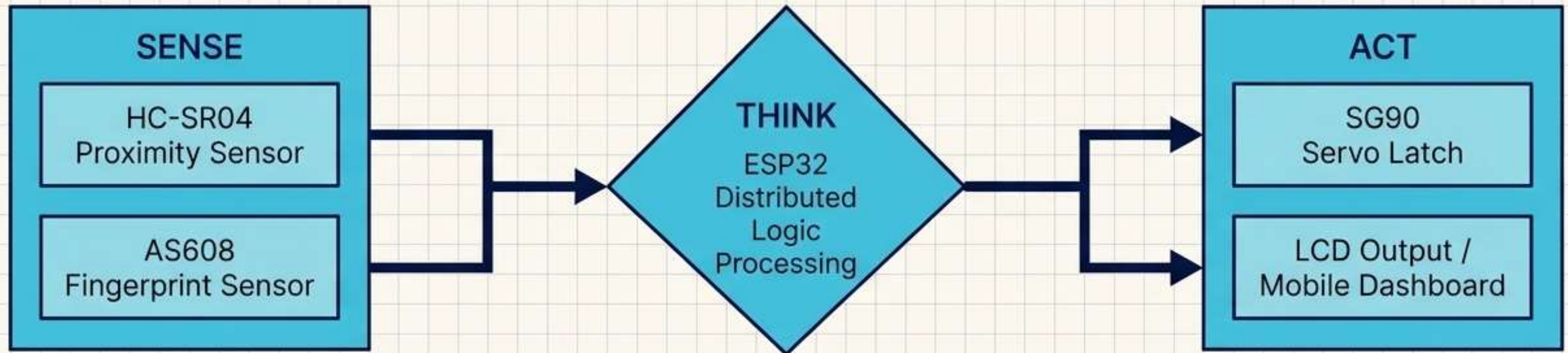
The Vulnerability Triad

	Key Vulnerability Physical keys and RFID cards can be stolen, lost, or shared, leading to unauthorized access.
	No Audit Trail Mechanical locks provide no data on who accessed the premises or at what time entry occurred.
	Remote Limits Standard locks cannot be monitored or controlled when the owner is away from the location.

SYSTEM OBJECTIVES



PROPOSED SOLUTION ARCHITECTURE



ANALYSIS & MODELING

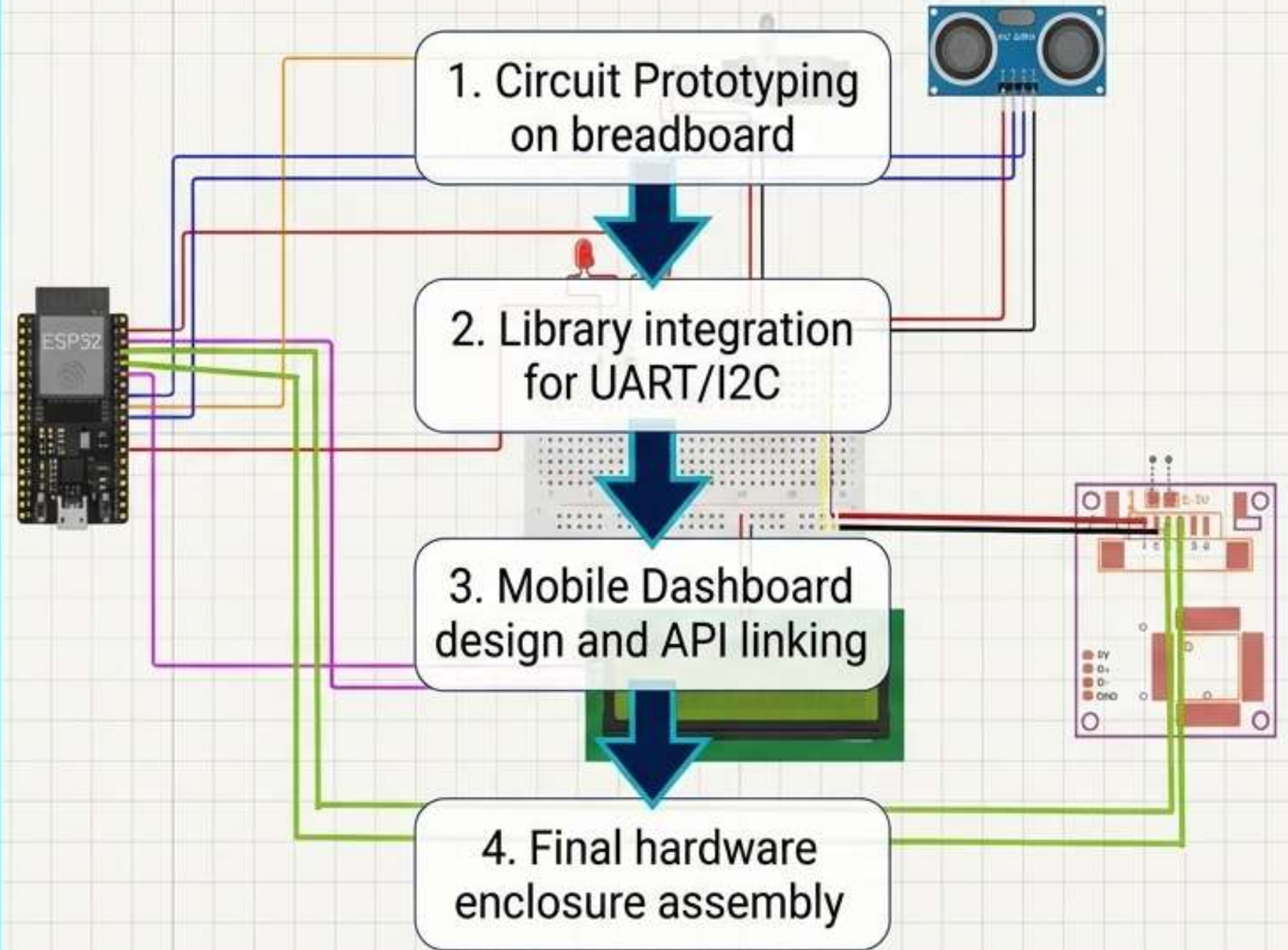
$$D = \frac{T * 0.0343}{2}$$

Where D = distance in cm and T = travel time of sound pulse in microseconds.

SYSTEM DESIGN & SOFTWARE STACK

- C++ Arduino Framework
- Blynk IoT for Mobile
- Preferences.h for NVS Storage
- Algorithms: Biometric Template Matching, PWM Latch Control

IMPLEMENTATION WORKFLOW



RESULTS & DIAGNOSTICS

✔ **<0.5s Latency**
(between authentication and actuation)

✔ **99% Accuracy**
(Biometric ID matching)

✔ **100% Stability**
(Mobile app synchronization)

✔ **Zero Data Loss**
(NVS storage retention)

DIAGNOSTIC PROFILE

Challenge: Initial ultrasonic noise causing false wake-ups.
Solution: Implementation of a software median filter.

ACCESS SYSTEM COMPARISON

Feature	Mechanical Key	RFID Card	Our Biometric IoT
Unique ID	No	Possible	Yes (Unforgeable)
Remote Status	None	Limited	Full Real-Time
Access Logs	Manual	Digital	Cloud-Automated

INNOVATION & FUTURE WORK



Facial Recognition

Integrating an ESP32-CAM for secondary authentication or intruder snapshots.



Solar Power

Developing a low-power deep-sleep mode powered by solar energy for remote gates.



Multi-Node

Expanding the network to manage multiple doors from a single enterprise app.

CONCLUSION

The **Smart Door Project** proves that highly secure, IoT-managed access is achievable with cost-effective embedded hardware.

The integration of biometrics and real-time cloud logging provides a scalable foundation for modern security systems.

